

CSA0711

26

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

**COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Code of Conduct:

I P. Sashank (19242534) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

P. Sashank

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Problem Understanding	20%	1.5	1.5	1.5	1.5	1.5
Method Selection	20%	1.5	1.5	1.5	1.5	1.5
Solution Process	25%	2	2	2	2	2
Accuracy of Calculation	20%	1.5	1.5	1.5	1.5	1.5
Interpretation of Results	15%	1/2	1/2	1/2	1/2	1

Total Marks

38
/40

[Signature]
Signature of the Faculty

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.
3. An organization has been allocated the IP block 172.16.0.0/16 for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or understands problem incorrectly

Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Problem understanding

A university has been assigned the IPv4 network 192.168.20.0 for internal communication. The administrator wants to divide this network into three separate subnets.

lab A $\rightarrow$ 126

lab B $\rightarrow$ 127

lab C $\rightarrow$ 128

The task is to determine:

- Network address of each subnet
- Valid host IP address range
- Broadcast address
- Number of usable host addresses

IPv4 subnetting formulas:

Number of Addresses

$$2^{(32 - \text{Prefix length})}$$

Usable Host Addresses

$$2^{(32 - \text{Prefix length})} - 2$$

Subnet sizes

126 $\rightarrow$ 64 addresses

127 $\rightarrow$ 32 addresses

128 $\rightarrow$ 16 addresses

Valid Hosts

192.168.20.1

to

192.168.20.62

$$64 - 2 = 62$$

Step 2: Allocate lab B (128)

The next available address after Lab A is

192.168.20.64

A/27 subnet contains

$$2^{32-27} = 32$$

addresses:

Network address

192.168.20.64/27

Address Range

192.168.20.64 - 192.168.20.95

Broadcast Address

192.168.20.95

Valid Hosts

192.168.20.65

to

192.168.20.94

Usable Hosts

$$32 - 2 = 30$$

Problem understanding

A network administrator has configured two computer system with the following IPv4 addresses and subnet mask:

- System 1: IP Address = 192.168.1.130, Subnet mask = 255.255.255.192
- System 2: IP Address = 192.168.1.190, Subnet mask = 255.255.255.192

The objective is to apply subnetting techniques to determine

- The subnet range for each system
- The network address
- The broadcast address
- The valid host IP address range
- Whether each IP address belongs to its respective subnet
- The number of usable host addresses available in each subnet

subnetting is an important networking technique that divides a large network into smaller logical networks.

method selection

The subnet mask 255.255.255.192 is equivalent to /26.

Formula for Total Addresses

$$2^{(32-26)} = 2^6 = 64 \text{ addresses}$$

Formula for Usable Hosts

$$2^6 - 2 = 62 \text{ usable hosts}$$

na

$$129 \leq 190 \leq 190$$

Usable Hosts

$$64 - 2 = 62$$

Usable Host Addresses = 62

Step 2: Determine the subnet of System 2

192.168.1.190

This address also falls within

192.168.1.128 - 192.168.1.191

Therefore,

Network Addresses = 192.168.1.128

Broadcast Address

192.168.1.191

Valid Host Range

192.168.1.129

to

192.168.1.190

Verification

The given IP is

192.168.1.190

Since

$$129 \leq 190 \leq 190$$

Usable Hosts

$$64 - 2 = 62$$

Usable Host Addresses = 62

R → 125

Sales → 126

IT → 127

Admin → 128

The objective is to:

- Assign suitable subnet addresses to each department
- Determine the network address, valid host IP range, and broadcast address for each subnet
- Calculate the number of usable host addresses available in each department
- Identify the remaining unused IP address range within the given 172.16.0.0/16 network block

Method selection

The network 172.16.0.0/16 provides a large address space containing:

$$2^{(32-16)} = 2^{16} = 65,536 \text{ IP addresses.}$$

The number of addresses and usable hosts for each subnet are calculated using:

Total Addresses

$$2^{(32 - \text{Prefix Length})}$$

Usable Host Addresses

$$2^{(32 - \text{Prefix Length})} - 2$$

Address Range

172.16.0.128 - 172.16.0.191

Valid Host Range

172.16.0.129 - 172.16.0.190

Broadcast Address

172.16.0.191

Step 3: Allocate 17 Departments (128)

The next available subnet begins at:

172.16.0.192

A /24 subnet provides:

$2^{32-24} = 32$ addresses

Usable Hosts:

32 - 2 = 30

Network Address

172.16.0.192 / 24

Address Range

172.16.0.192 - 172.16.0.223

Valid Host Range

172.16.0.193 - 172.16.0.222

Broadcast Address

172.16.0.223

method selection

Network Address: 172.16.100/24

subnet mask: 255.255.255.224

Formula 1: Number of subnet.

$$24 - 24 = 0$$

$$2^0 = 1 \text{ subnet}$$

$$2^{(32-24)} = 2^8 = 32 \text{ addresses}$$

$$32 - 2 = 30 \text{ usable hosts}$$

$$256 - 224 = 32$$

The subnet addresses increase by 32.

Determine the Subnet for IP Address 172.16.10.78

172.16.10.78

This address falls within the range

172.16.10.64 - 172.16.10.95

subnet: 172.16.10.64/27

Network Address

172.16.10.64

Broadcast

172.16.10.95

Valid Host Range

172.16.10.65 - 172.16.10.94

Problem Understanding

A Network uses the IPv6 address:

2001:0db8:0000:0000:0000:4500:0042:8321

The task is to Apply IPv6-Addressing back to the

full notation.

Determine the Network Prefix using a /64

Prefix length.

Calculate the total number of possible host addresses available within the network.

IPv6 Address Compression follows these:

1) Remove the leading zeros from each hex block.

2) Replace one consecutive sequence of a block with.

F8 Expansion:

Restore omitted zero blocks.

Add leading zero to make each

Compressed IPv6 Address

2001:db8::ff00:42:8329

This is the shortest valid representation of the given IPv6 address

Step 3:- Expand the compressed address

Starting with

2001:db8::ff00:42:8329

The symbol :: represents the omitted three zero blocks.

After restoring the missing blocks and adding leading zeros:

2001:0db8:0000:0000:0000:ff00:0000:428329

This matches the original IPv6 address